

SIMATS ENGINEERING
ASSESSMENT TOOL 3
Mock Interview

COURSE NAME: COMPUTER VISION WITH OPENCV
COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO5: Understand video processing - motion computation - 3D vision - geometry. (BTL 6)

Assessment Tool Weightage: 10%

Code of Conduct:

*I _Chalamala Sai Harshitha (192424346)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: C. Sai Harshitha

Mock Interview Questions

1. Face & Expression Recognition

A face detection system must run on low-power devices. Design an optimized pipeline and justify your model selection.

Introduction

A face and expression recognition system on a low-power device must provide accurate results while using limited computational resources. The pipeline should therefore minimize unnecessary processing and use lightweight models.

1. Camera Input

- Capture frames continuously using the mobile camera.
- Process the frames efficiently to reduce unnecessary computation.
- Avoid using excessively high-resolution input.

2. Frame Preprocessing

- Resize the captured frame to a suitable resolution.
- Convert the frame into the required format.
- Normalize the input if required by the model.
- Smaller images reduce processing time and memory usage.

3. Face Detection

- Detect faces present in the input frame.
- Draw a bounding box around each detected face.
- A lightweight face detection model should be preferred.
- Multiple faces should be handled independently.

4. Face ROI Extraction

- Extract the **Region of Interest (ROI)** for each detected face.
- Process only the face region instead of the complete image.
- Resize the ROI according to the classifier's input requirements.
- This reduces unnecessary background processing.

5. Facial Feature Extraction

- Extract important visual features from the face ROI.
- Features may include patterns around the eyes, eyebrows, nose, and mouth.
- A lightweight CNN can be used to learn these facial patterns.

6. Expression Classification

The extracted features are given to an expression classifier.

The system can classify the expression into:

- **Happy**
- **Sad**
- **Neutral**

The predicted expression is associated with the corresponding detected face.

7. Result Display

- Draw the face bounding box on the original frame.
- Display the predicted expression near the detected face.
- Update the result continuously as new frames are processed.

Model Selection

A **lightweight CNN-based model** is suitable for this application because the system must operate on a low-power device.

Reasons for Selecting a Lightweight Model

Requirement	Benefit
Low memory	Smaller number of model parameters
Low computation	Fewer operations during inference
Fast response	Faster prediction
Low power	Reduced processor workload
Real-time operation	Faster processing of camera frames

A large deep learning model may provide high accuracy but can require more memory and processing power. Therefore, a lightweight model provides a better balance between **accuracy** and **computational efficiency** for this application.

Optimization Techniques

1. Frame Resizing

Reduce the input resolution before processing to decrease computational workload.

2. ROI-Based Processing

Process only detected face regions instead of the complete image.

3. Lightweight Model

Use a compact face detector and expression classifier suitable for mobile or embedded devices.

4. Efficient Frame Processing

Avoid unnecessary processing of every frame at maximum resolution.

5. Memory Optimization

Avoid unnecessary copies of frames and intermediate images.

6. Fast Inference

Use an optimized model so that predictions can be generated with minimum delay.

Real-Time Processing

For real-time operation, the pipeline should process frames in the following order:

Capture → Resize → Detect → Extract ROI → Classify → Display

The processing time should be kept low at each stage. If required, face detection can be performed less frequently while tracking previously detected faces between detection steps.

Advantages

- Suitable for low-power devices.
- Requires less memory.
- Reduces computational workload.
- Provides faster inference.
- Supports real-time processing.
- Can handle multiple detected faces.
- Reduces unnecessary background processing.
- Provides a modular and easily optimizable pipeline.

Conclusion

The proposed pipeline provides an efficient approach for face and expression recognition on low-power devices. Frame resizing, ROI extraction, lightweight models, and efficient frame processing reduce computational requirements. A lightweight CNN-based model is therefore appropriate because it provides a suitable balance between **recognition performance, processing speed, memory usage, and power consumption.**

2. OCR & Text Processing

You need to detect handwritten text in noisy images. Design a solution and justify your choices.

Introduction

Handwritten text in noisy images can be difficult to recognize because of blur, uneven lighting, background noise, broken characters, and variations in handwriting. An effective OCR pipeline should therefore first improve the image quality and then perform text detection and recognition.

1. Image Acquisition

- Read the input image containing handwritten text.
- Check the image quality and resolution.
- Convert the image into a suitable format for further processing.

2. Grayscale Conversion

- Convert the color image into a grayscale image.
- This reduces the image information from three color channels to a single intensity channel.
- It simplifies subsequent image processing.

3. Noise Removal

Noise should be removed before OCR processing.

Possible filtering methods include:

- **Median filtering** – useful for removing salt-and-pepper noise.
- **Gaussian filtering** – useful for reducing general image noise.
- **Bilateral filtering** – can reduce noise while preserving important edges.

The filter should be selected according to the type of noise present in the input image.

4. Image Enhancement

- Improve contrast between the handwritten text and the background.
- Use contrast enhancement when the text is faint.
- Correct uneven illumination if required.
- This makes the handwritten characters easier to detect.

5. Thresholding

- Convert the enhanced grayscale image into a binary image.
- Separate the handwritten text from the background.
- **Adaptive thresholding** is suitable when the image contains uneven lighting or a non-uniform background.

6. Morphological Processing

Morphological operations can improve the binary text image.

- **Opening** can remove small unwanted noise.
- **Closing** can fill small gaps in characters.
- These operations help produce cleaner handwritten characters before OCR.

7. Text Detection and Segmentation

- Identify regions containing handwritten text.
- Detect connected components or contours.
- Separate individual words or characters when required.
- Maintain the correct reading order.

8. OCR Processing

The cleaned text regions are passed to an OCR engine.

The OCR stage:

- Analyzes the handwritten characters.
- Recognizes the characters or words.
- Converts the visual information into machine-readable text.

For handwritten text, a recognition model trained specifically for handwriting should be preferred over a general printed-text OCR system.

9. Output Processing

- Combine the recognized characters or words.
- Preserve the correct word and line order.
- Remove obvious unwanted OCR artifacts when possible.
- Display or store the final recognized text.

Justification of Choices

Stage	Choice	Justification
Preprocessing	Grayscale	Reduces image complexity
Noise removal	Median/Gaussian filter	Reduces unwanted noise
Enhancement	Contrast adjustment	Improves text visibility
Thresholding	Adaptive thresholding	Handles uneven backgrounds
Morphology	Opening and closing	Removes noise and fills gaps
Segmentation	Contours/connected components	Identifies text regions
OCR	Handwriting-aware OCR	Better suited to handwritten characters

Handling Noisy Images

The pipeline should adapt according to the type of noise:

- **Salt-and-pepper noise:** Use median filtering.
- **Gaussian noise:** Use Gaussian filtering.
- **Uneven lighting:** Use adaptive thresholding.
- **Broken characters:** Use morphological closing.
- **Small unwanted regions:** Use morphological opening.
- **Low contrast:** Apply contrast enhancement.

Advantages

- Improves the quality of noisy handwritten images.
- Reduces unwanted background information.
- Makes handwritten characters easier to detect.
- Provides a structured OCR pipeline.
- Can handle variations in image quality.
- Produces machine-readable text from handwritten input.

Conclusion

The proposed OCR pipeline first improves the quality of the noisy image through grayscale conversion, filtering, enhancement, thresholding, and morphological processing. The cleaned image is then segmented into text regions and passed to a handwriting-aware OCR system. Adaptive thresholding and appropriate filtering are particularly useful for handling noisy and unevenly illuminated images. This structured approach improves the reliability of handwritten text detection and recognition.

3. Application-Oriented Design

Design a document scanning system that enhances readability automatically. Justify each stage of your pipeline.

Introduction

A document scanning system should convert a captured document image into a clear and readable digital document. Since scanned images may contain skew, shadows, noise, poor lighting, and low contrast, the system should automatically process the image before producing the final output.

1. Image Acquisition

- Capture the document using a camera or scanner.
- Read the captured image into the system.
- Ensure that the complete document is available in the input frame.

Justification:

A clear input image provides better results during document detection and enhancement.

2. Document Detection

- Detect the boundaries of the document.
- Identify the four corners or boundary points.
- Separate the document from the surrounding background.

Justification:

Only the document region should be processed. Removing the background improves the quality of the final scanned document.

3. Perspective Correction

- Use the detected corner points to correct the document's perspective.
- Apply a perspective transformation.
- Convert the document into a rectangular, front-facing view.

Justification:

Documents captured at an angle may appear distorted. Perspective correction makes the text properly aligned and improves readability.

4. Grayscale Conversion

- Convert the corrected document image into grayscale.
- Reduce the image to intensity information.

Justification:

Grayscale processing reduces computational complexity and is sufficient for most document enhancement operations.

5. Noise Removal

- Apply an appropriate smoothing filter.
- Remove unwanted noise and small image artifacts.
- Preserve important text edges.

Justification:

Noise can interfere with text detection and thresholding. Removing it produces a cleaner document image.

6. Contrast Enhancement

- Improve the contrast between the text and background.
- Enhance faint or poorly visible characters.
- Correct uneven illumination when required.

Justification:

Better contrast makes text more distinguishable from the document background.

7. Thresholding

- Convert the enhanced grayscale image into a binary image.
- Separate foreground text from the background.
- Use adaptive thresholding when lighting is uneven.

Justification:

Thresholding produces a document with clearer text and background separation.

8. Morphological Processing

Apply suitable morphological operations such as:

- **Opening:** Removes small unwanted noise.
- **Closing:** Fills small gaps in characters.

Justification:

Morphological processing improves the structure of text and removes remaining small imperfections.

9. Output Generation

- Generate the enhanced document image.
- Display or save the final scanned document.
- Preserve the document content while improving readability.

Justification:

The final output should be clean, properly aligned, and easier to read than the original captured image.

Pipeline Summary

Stage	Main Operation	Purpose
1	Image Acquisition	Capture document
2	Document Detection	Locate document boundaries
3	Perspective Correction	Remove geometric distortion
4	Grayscale Conversion	Simplify processing
5	Noise Removal	Reduce unwanted artifacts
6	Contrast Enhancement	Improve text visibility
7	Thresholding	Separate text and background
8	Morphological Processing	Improve text structure
9	Output Generation	Produce readable scanned document

Advantages

- Automatically improves document readability.
- Corrects perspective distortion.
- Reduces image noise.
- Enhances text visibility.
- Handles uneven lighting using adaptive processing.
- Produces a clean digital document.
- Reduces the need for manual image editing.

Conclusion

The proposed document scanning pipeline automatically converts a captured document into a clearer and more readable digital image. Document detection and perspective correction ensure proper alignment, while grayscale conversion, noise removal, contrast enhancement, thresholding, and morphological processing improve the visual quality of the document. Each stage has a specific purpose and contributes to producing a clean, readable, and practically useful scanned document.

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately